

# INS Statistical Survey Collection Platform

## Technology Stack

### Frontend

- Angular 20
- Angular Material
- Bootstrap 5
- ngx-translate
- ApexCharts

### Backend

- FastAPI
- Pydantic
- SQLAlchemy 2
- Alembic

### Database

- MySQL 8

### Authentication & Security

- JWT (python-jose)
- Passlib + Bcrypt

### Background Processing

- Redis
- Celery
- Celery Beat

### Email Service

- Brevo

### File Handling & Reporting

- OpenPyXL
- Pandas
- WeasyPrint

### Infrastructure

- Docker
- Docker Compose
- Nginx (future deployment)

# Development Tools

- VS Code
- Postman
- MySQL Workbench
- Git
- GitHub

# Architecture Overview

Angular + Angular Material + Bootstrap + ngx-translate → FastAPI + Pydantic + SQLAlchemy + Alembic → MySQL → Redis + Celery + Celery Beat → Brevo

## Recommended Learning Order

- Git & GitHub
- Docker
- MySQL
- SQLAlchemy
- FastAPI
- JWT Authentication
- Angular
- Angular Material
- Redis
- Celery
- Brevo
- Docker Compose